$$L_x = I\,\omega_x$$

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

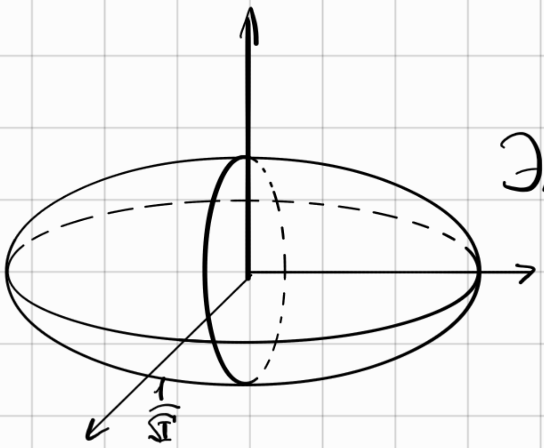
тензор инерции

$$I_{xx} = \int (y^2 + z^2)\,dm$$

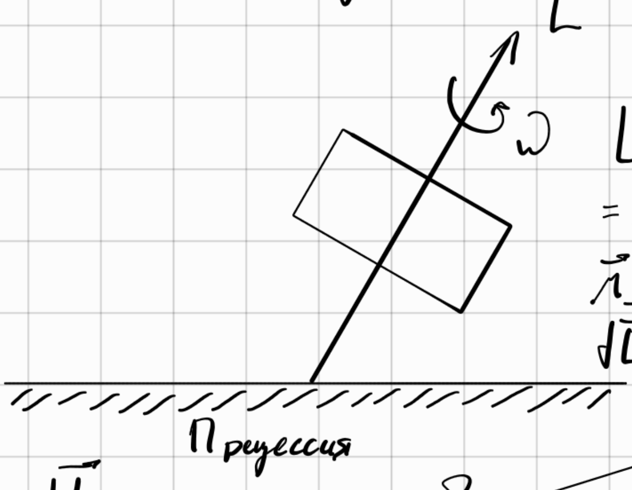
$$I_{xy} = -\int xy\,dm = I_{yx}$$

Тензор - сим. матр. => $\exists$ главные оси:

$$\begin{pmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{pmatrix}$$



Эллипсоид инерции



$L \gg dL =$

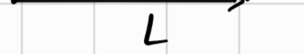
$= Mdt \;\forall dt$

$\vec M \perp \vec L$

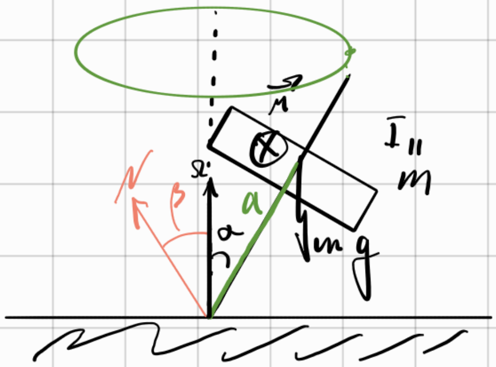
$d\vec L \perp \vec L$

Прецессия

$$\frac{d\vec L}{dt} = \vec M = [\vec\Omega \times \vec L]$$



11.1



$N_y = mg$

$M = mga\sin\alpha$

$N_x = m\Omega^2 a\sin\alpha$

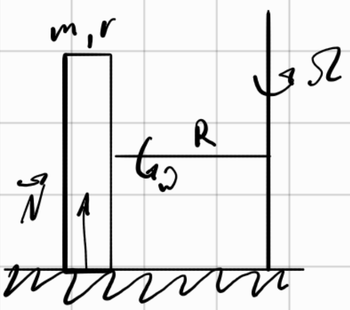
$tg\beta = \frac{N_x}{N_y} = \frac{\Omega^2 a\sin\alpha}{g}$

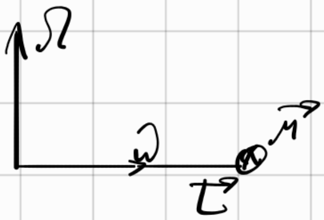
$L = I_\parallel\omega$

$M = \Omega\cdot I_\parallel\omega\cdot\sin\alpha$

$\Omega = \frac{mga}{I_\parallel\omega}$

11.14





$M = FR$

$\parallel$

$\Omega\cdot L_c = \Omega\frac{mR^2}{2}\omega$ $\qquad\omega = \Omega\frac{R}{r}$

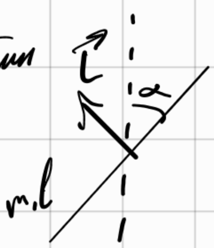
$F = \frac{m\Omega^2 r}{2}$

$N = mg + \frac{m\Omega^2 r}{2}$

11.24

$n = 3000\ \frac{1}{мин}$

$\alpha = 89{,}9^\circ$



$dm = \frac{m}{l}dx$

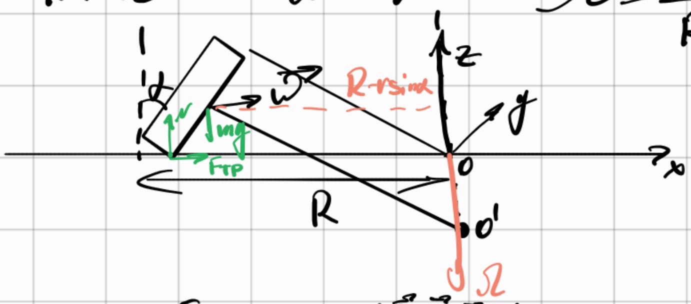
$dL = x\,dp = x\,dm\,\omega\,x\sin\alpha$

$$L = \int_{-\frac{l}{2}}^{\frac{l}{2}}\frac{m}{l}\omega\sin\alpha\,x^2dx = \frac{m}{l}\omega\sin\alpha\,\frac{x^3}{3}\Big|_{-\frac{l}{2}}^{\frac{l}{2}} =$$

$$= \frac{ml^2}{12}\omega^2\sin\alpha$$

$$M = \omega L\cdot\sin\left(\frac{\pi}{2}-\alpha\right) = \frac{ML^2}{12}\omega^2\overbrace{\sin\alpha}^{\approx 1}\overbrace{\sin\left(\frac{\pi}{2}-\alpha\right)}^{\approx\frac{\pi}{2}-\alpha}$$

11.18 $\omega = \frac{v}{r}$ $\Omega = \frac{v}{R}$



$$\vec{M} = [\vec{\Omega} \times \vec{L}] = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\Omega \\ -L_x & 0 & L_z \end{vmatrix} = \vec{j}\,\Omega L_x$$

$$m\Omega^2 (R - r\sin\alpha) = F_{TP}$$

$$N = mg$$

$$M_N = mgR$$

$$M_{mg} = -mg(R - r\sin\alpha)$$

$$M_{TP} = m\Omega^2 (R - r\sin\alpha) r_{00'} = m\Omega^2 (R - r\sin\alpha)\frac{R\sin\alpha - r}{\cos\alpha}$$

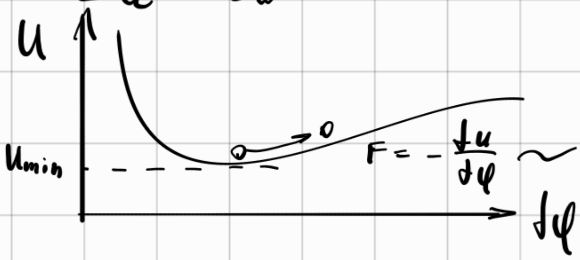
$$\Omega L_x = \frac{(R - r\sin\alpha)(R\sin\alpha - r)}{\cos\alpha} m\Omega^2 - (R - r\sin\alpha)mg + mgR$$

$$\vec{L} = \vec{L}_{\parallel} + \vec{L}_{\perp}$$

$$L_{\parallel} = I_{\parallel}\,\omega_{\parallel} = mr^2(\omega + \Omega_{\parallel})$$

$$L_{\perp} = I_{\perp}\,\omega_{\perp} = \left(\frac{mr^2}{2} + m\left(\frac{(R - r\sin\alpha)^2}{\cos\alpha}\right)\right)\Omega_{\perp}$$

$$L_{\Omega} \sim L_{\omega}$$



$$F = -\frac{dU}{d\varphi} \sim -K\Delta\varphi$$

543

$$F = K(l - l_0)^2$$

$$mg = K\Delta l_0^2$$

$$\Delta l_0 = \sqrt{\frac{mg}{K}}$$

$$\vec{F}_y = K(\Delta l_0 + x)^2$$

$$m\ddot{x} = mg - K(\Delta l_0 + x)^2$$

$$m\ddot{x} = -2k\Delta l_0 x$$

$$\ddot{x} + \frac{2k\Delta l_0}{m} x = 0$$

$$\omega = \sqrt{\frac{2k\Delta l_0}{m}} = \sqrt[4]{\frac{4kg}{m}}$$

$$\Pi = \frac{A\varphi^2}{2}$$

$$K = B\frac{\dot{\varphi}^2}{2}$$

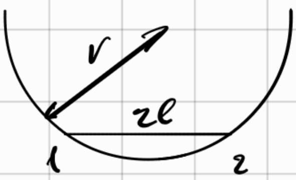
$$E = A\frac{\varphi^2}{2} + B\frac{\dot{\varphi}^2}{2}$$

$$\dot{E} = 0 = A\varphi\dot{\varphi} + B\dot{\varphi}\ddot{\varphi}$$

$$0 = A\varphi + B\ddot{\varphi}$$

$$\ddot{\varphi} + \frac{A}{B}\varphi = 0 \quad \omega = \sqrt{\frac{A}{B}} \quad T = 2\pi\sqrt{\frac{B}{A}}$$

5.22



1) ⊥ рис

$$\rho = \sqrt{r^2 - l^2}$$

$$\Pi = 2mg(\rho - \rho\cos\varphi) \approx \overbrace{2mg\rho}^{A}\frac{\varphi^2}{2}$$

$$K = \overbrace{2m\frac{\dot{\varphi}^2}{2}\rho}^{B}$$

$$T = 2\pi\sqrt{\frac{B}{A}} = 2\pi\sqrt{\frac{2m\rho^2}{2mg\rho}} = 2\pi\frac{\sqrt[4]{r^2 - l^2}}{\sqrt{g}}$$

2)

$$\Pi = 2mg\rho\frac{\varphi^2}{2}$$

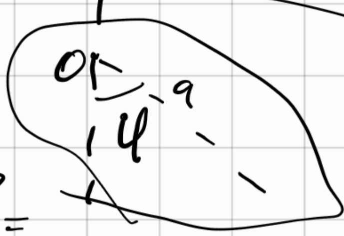
$$K = 2mr^2\frac{\dot{\varphi}^2}{2}$$

$$T = 2\pi\sqrt{\frac{r}{\rho\sqrt{r^2 - l^2}}}$$

Физ. маятник

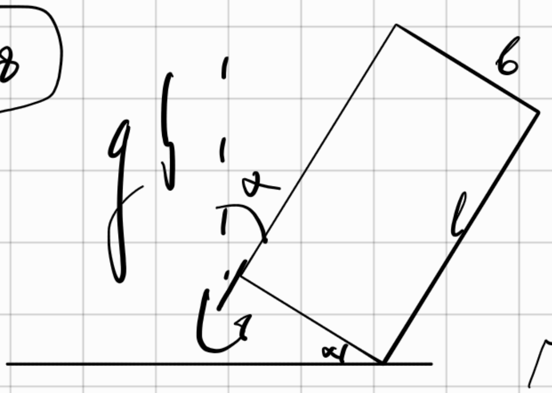
$$I\ddot{\varphi} = -mga\sin\varphi$$

$$\ddot{\varphi} + \frac{mga}{I}\varphi = 0 \qquad T = 2\pi\sqrt{\frac{I}{mga}} =$$



$= 2\pi\sqrt{\frac{l_{пр}^2}{g}}$ $\quad l_{пр} = \frac{I}{ma}$

1048



$T - ?$

$\frac{b}{2}(1-\cos\varphi)$

$\frac{b}{2}\sin\alpha(1-\cos\varphi)$

$П \approx mg\frac{b}{2}\sin\alpha\,\frac{\varphi^2}{2}$

$K = I\frac{\dot\varphi^2}{2};\ I = \frac{mb^2}{3}$

$T = 2\pi\sqrt{\frac{2b}{3g\sin\alpha}}$